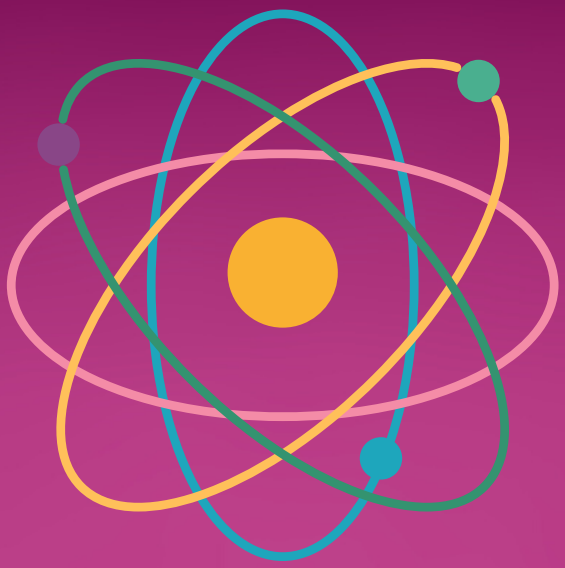
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
path to success



PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Thermodynamics-01

English Medium

EXERCISE # S-I

BASIC

1. Find the number of quantities which are intensive properties among the following.

Temperature, Boiling point, Melting point, Pressure , Density , Viscosity , Surface Tension, Refractive Index, Molar volume, Molar Enthalpy, Specific heat, Specific volume , Mass , Volume, Number of moles , Internal energy, Enthalpy , Entropy , Concentration, Dipole moment, pH, Gas constant, Vapour pressure, Specific gravity, E.M.F. of the dry cell, Molarity, Molality .

TE0112

2. Find the number of following functions which are macroscopic properties.

Pressure, Volume, Enthalpy, Work, Heat, Gibbs energy, temperature, Internal energy, Entropy.

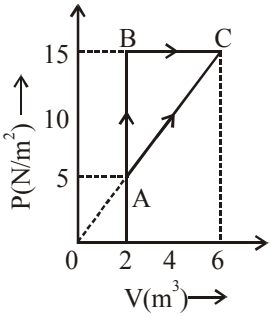
3. The gas is cooled such that it loses 65 J of heat. The gas contracts as it cools and work done on the system is equal to 20 J. What are q, w and ΔU (in Joule) ?

TE0113

4. Find the work done (in atm-litre) when 1 mole of gas expands from 1 litre to 5 litre against constant of 1 atm pressure.

TE0002

5. The given figure shows a change of state A to state C by two paths ABC and AC for an ideal gas. Calculate the :



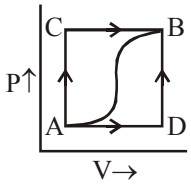
(a) Internal energy at C if the internal energy (in Joule) of gas at A is 10 J and amount of heat supplied to change its state to C through the path AC is 200J.

TE0021

(b) Amount of heat (in joule) supplied to the gas to go from A to B, if internal energy change of gas from A to B is 10 J.

TE0022

6. When a system is taken from state A to state B along the path ACB, 80J of heat flows into the system and the system does 30J of work. Find (in joule)



(a) How much heat flows into the system along path ADB if the work done by the system is 10J ?

TE0006

(b) When the system is returned from state B to A along the curved path,the work done on the system is 20J. Does the system absorb or liberate heat, and how much ?

TE0007

(c) If $U_D - U_A = 40J$, find the heat absorbed in the process AD and DB if work done by system in ADB is 10 Joule.

TE0008

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

7. Find total degree of freedom and break up as translational, rotational or vibrational degree of freedom in following cases.

(i) CO_2 (ii) SO_2 (iii) He (iv) NH_3

TE0004

8. 1 mol of an ideal diatomic gas is expanded by heating from state-I (10 litre, 500K) to (20 litre, 800K) against constant pressure 2 bar. Find heat supplied in joule. (Given R = 25/3 J/mole/K)

TE0114

9. Enthalpy change in calorie for 1 mol of diatomic ideal gas under going the state change (2 atm, 300 K) to (3 atm, 360 K) will be ?

TE0115

10. One mole of an ideal monoatomic gas undergoes expansion along a straight line on P–V curve from initial state A(3L, 8 atm) to final state B(7.5 L, 2 atm). Calculate q for the above process in L atm.

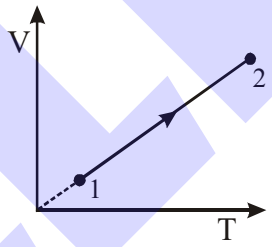
TE0026

11. Molar heat capacity of an ideal gas at constant volume is given by $C_V = (16.5 + 10^{-2}T)\ \frac{J}{K-mol}$.

If 2 moles of this ideal gas are heated at constant volume from 300K to 400K. Then the change in internal energy (in kJ) will be approximately.

TE0016

12. Volume vs Temperature graph of 3 moles of O_2 gas is as shown in figure. The ratio of heat absorbed & the work done by the gas in process 1-2 is -



TE0116

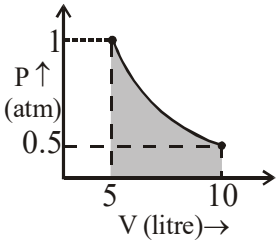
13. If 1.0 kJ of heat is added to 1.2 L of a gas in a cylinder at constant pressure of 1 bar, the volume increases to 1.5L. Calculate ΔU and ΔH of the process in Joules.

TE0117

14. When 1 mole of ice melt at 0°C and at constant pressure of 1 atm, 1440 calories of heat are absorbed by the system. The molar volumes of ice and water are 0.0196 and 0.0180 litre respectively. Calculate ΔH and ΔU (in calori) for the reaction. **TE0118**

15. Find the work done (in kJ) when one mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 27°C. (ln5 = 1.6) **TE0119**

16. Shaded area in the graph will be (in atm-L) : (ln2 = 0.7)



TE0120

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol-TD-01-TC-TD-02 & 1.5_Eng (PC)\Eng\1-Thermodynamic-01\02_Ex.p65

17. An ideal gas expands isothermally from 10^{-3} m^3 to 10^{-2} m^3 at 300 K against a constant pressure of 10^5 Nm^{-2}. Calculate work in joule. **TE0121**

18. Five moles of an ideal gas at 300 K, expanded isothermally from an initial pressure of 4 atm to a final pressure of 1 atm against a constant external pressure of 1 atm. Calculate q, w, ΔU & ΔH (in atm-litre). Calculate the corresponding values of (q, w, DU & DH) (in atm-litre) if the above process is carried out reversibly. (ln2 = 0.7) **TE0122**

19. An ideal gas undergoes expansion from A(10 atm, 1 litre) to B(1 atm, 10 litre), first against 5 atm and then against 1 atm, isothermally. Calculate the amount of heat absorbed (in litre-atm) **TE0009**

20. 2 mole of an ideal gas undergoes isothermal compression along three different paths if it's initial state is (2 bar, 4 litre) in all three process.

(i) Reversible compression from till 20 bar.

(ii) A single stage compression against a constant external pressure of 20 bar

(iii) A two stage compression consisting initially of compression against a constant external pressure of 10 bar until $P_{gas} = P_{ext}$, followed by compression against a constant pressure of 20 bar until $P_{gas} = P_{ext}$.

Calculate the work (in bar. L) for each of these processes and for which of the irreversible processes is the magnitude of the work greater ? [Given : R = 0.08 bar. L/mole.K] **TE0011**

21. A gas is expanded in adiabatic process from 2L to 5L against constant external pressure of 1 bar then internal energy change in bar litre will be : **TE0123**

22. 2 moles of an ideal gas is compressed from (1 bar, 2L) to 2 bar isothermally. Calculate magnitude of minimum possible work involved in the change (in Joules). (Given : 1 bar L = 100 J) (ln 2 = 0.7) **TE0010**

23. Two mole of ideal diatomic gas ($C_{V,m} = 5/2$ R) at 300K and 5 atm expanded irreversibly & adiabatically to a final pressure of 2 atm against a constant pressure of 1 atm. Calculate q, w, ΔH & ΔU. (in calories) **TE0013**

24. Calculate the work done in calories by system in an irreversible (single step) adiabatic expansion of 1 mole of a polyatomic gas ($\gamma = 4/3$) from 400 K and pressure 16 atm to 1 atm. Also calculate work done in kcal if process done reversibly and adiabatically. **TE0124**

25. 1 mole of CO_2 gas at 300 K is expanded under reversible adiabatic condition such that its volume becomes 27 times.

(a) What is the final temperature (in K)?

(b) What is work done (in kJ) ?

Given γ = 1.33 and C_V = 25.08 J $mol^{-1}K^{-1}$ for CO_2. **TE0015**

node06\B0BC-BD\Kota\JEE(Advanced)\Enfuse\Chem\Module\4TD-01-TC-TD-02 &1.Eq.-IPC\Eng\1-Thermodynamic-01\02_Ex.p65

26. Three moles of an ideal gas at 200 K and 2.0 atm pressure undergoes reversible adiabatic compression until the temperature becomes 250 K. For the gas C_V is 27.5 JK^{-1} mol^{-1} in this temperature range. Calculate q, w, ΔU, ΔH (in kJ) and final V (in dm^3) and final P (in atm).

Given $\left(\frac{5}{4}\right)^{1/0.3} = 2.1$, $\left(\frac{5}{4}\right)^{35.8/8.3} = 2.61$

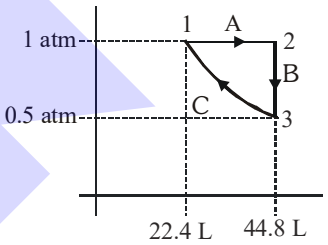
[R = 8.3 Joule/mol.K or R = 0.0821 atm litre/mol.K] **TE0017**

27. A monoatomic ideal gas undergoing irreversible adiabatic compression from 4L to 1L against 1 bar pressure. Calculate the enthalpy change of gas (in J) . (Given: 1 bar-L = 100 J)

TE0012

28. One mole of an ideal monoatomic gas is put through reversible path as shown in figure. Fill in the blank in the tables given below. Calculate q, w, ΔH & ΔU in cal. (ln2 = 0.7)

Table-1			
State	P (atm)	V (Lit.)	T(K)
1			
2			
3			

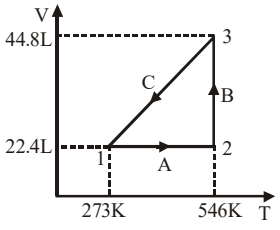


Step	Name of process	q	w	ΔU	ΔH
A					
B					
C					
	cyclic				

TE0125

29. One mole of an ideal monoatomic gas is carried through the reversible cycle of the given figure consisting of step A, B and C and involving state 1, 2 and 3. Fill in the blank space in the table given below assuming reversible steps. Calculate q, w, ΔH & ΔU in cal. (ln2 = 0.7)

Table-1			
State	P (atm)	V (Lit.)	T(K)
1			
2			
3			

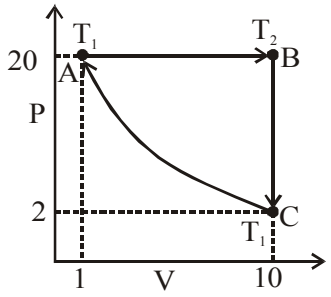


Step	Name of process	q	w	ΔU	ΔH
A					
B					
C					
overall					

TE0126

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1.5_Eng (PC)\Eng\1-Thermodynamic-01\02_Ex.p65

30. One mole of a perfect monoatomic gas is put through a cycle consisting of the following three reversible steps :



(CA) Isothermal compression from 2 atm and 10 litres to 20 atm and 1 litre.

(AB) Isobaric expansion to return the gas to the original volume of 10 litres with T going from T_1 to T_2. (BC) Cooling at constant volume to bring the gas to the original pressure and temperature. The steps are shown schematically in the figure shown.

(a) Calculate T_1 and T_2 (in K)

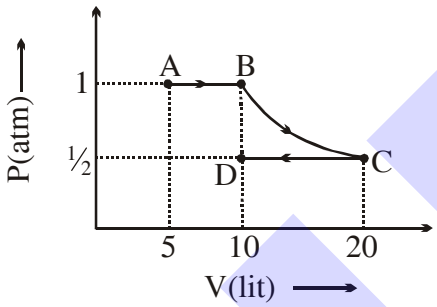
(b) Calculate ΔU, q and w (in calories) for the cycle.

[Give all answers as nearest integer]

[Given : ln10 = 2.3, R = 0.0821 atm.litre/mol-K or R = 2 cal/mol.K]

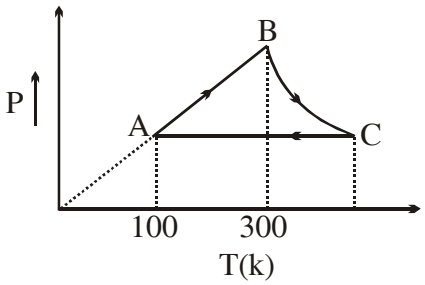
TE0127

31.



Calculate magnitude of total work done (in atm.lit) for the above process ABCD involving a monoatomic ideal gas. **[Given : *l*n 2 = 0.7]** **TE0024**

32. Calculate the net work done in the following cycle for one mol of an ideal gas (in calorie), where in process BC, PT = constant. (R = 2cal/mol-K).



TE0023

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4TD-01-TC-TD-02 & 1.Eq.-IPC\Eng\1-thermodynamic-01\02_Ex.p65

33. $\frac{1}{22.4}$ mol of an ideal monoatomic gas undergoes a reversible process for which $PV^2 = C$. The gas is expanded from initial volume of 1 L to final volume of 2 L starting from initial temperature of 273 K. Find the heat exchanged q during the process. Express your answer in litre atm.

$R = \frac{22.4}{273}$ L atm mole^{-1} degree^{-1}.

TE0124

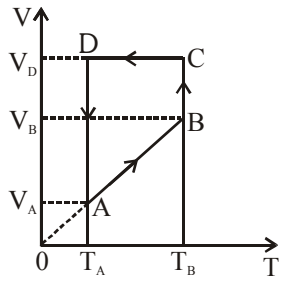
34. For 1 mole of ideal monoatomic gas if $\frac{P}{V^2}$ = constant and initial temperature is 100 K. If gas is expanded from 1 L to 2 L then find (a) heat capacity of the process (in cal/K-mol) (b) total heat absorbed (in cal) (c) work (in cal) (d) (in cal) change in internal energy.

(Assume $R = 0.0821 \frac{L - atm}{mole - k}$, 1 L-atm = 24 cal) **TE0025**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1.Eq_(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

EXERCISE # S-II

1. A monoatomic ideal gas of two moles is taken through a reversible cyclic process starting from A as shown in figure. The volume ratios are $\frac{V_B}{V_A} = 2$ and $\frac{V_D}{V_A} = 4$. If the temperature T_A at A is 27°C, calculate:



(a) The temperature of the gas at point B in kelvin.

TE0030

(b) Heat absorbed or released by the gas in each process in kcal.

TE0031

(c) The total work done by the gas during complete cycle. (ln2 = 0.7) in kcal

TE0032

2. One mole of an ideal gas (not necessarily monoatomic) is subjected to the following sequence of steps.

(a) It is heated at constant volume from 298 K to 373 K

(b) It is expanded freely into a vacuum to double volume at 373 K.

(c) It is cooled reversibly at constant pressure to 298K.

Calculate q, w, ΔU and ΔH for the overall process in joule.

TE0033

3. A real gas follows equation $[PV = nRT - \frac{an^2}{V}]$. If 1 mole of this real gas is reversibly & isothermally expanded from 1 litre to 2 litre of 300 K then calculate work in bar litre for the process ?

$[R = 0.08 \ \frac{bar - litre}{mol - K}$, ln2 = 0.7 and a = $\frac{1}{4}$ bar litre^2/mole$^2]$

TE0034

4. 3 dm^3 of an ideal monoatomic gas at 600 K and 32 bar expands until pressure of the gas is 1bar Calculate q, w, ΔU and ΔH for the process in cal if the expansion is :

(Use : R = 0.08 bar litre/mol-K = 2 cal / mol-K)

(i) Isothermal and reversible

TE0035

(ii) Adiabatic and reversible

TE0036

(iii) Isothermal and adiabatic

TE0037

(iv) Against 1 bar and adiabatic

TE0038

(v) Against 1 bar and isothermal.

TE0039

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4TD-01-TC-TD-02 &1.Eq.-(PC)\Eng\1-thermodynamic-01\02_Ex.p65

5. Pressure over 1000 mL of a liquid is gradually increased from 1 bar to 1001 bar under adiabatic conditions. If the final volume of the liquid is 990 mL, calculate ΔU and ΔH in joule of the process, assuming linear variation of volume with pressure.

TE0040

6. At 500 kilobar pressure, density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value $|\Delta H - \Delta U|$ (kJ/mole) for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T' :

TE0041

7. 2 mole of ideal monoatomic gas was subjected to reversible adiabatic compression from initial state of P = 1 atm and T = 300 K till the pressure is $4\sqrt{2}$ atm and temperature is TK. The gas is then subjected to reversible isothermal expansion at TK till the internal pressure is one atm. The gas is now isobarically cooled to attain initial state. Find W_{Net} (in calorie) for whole process.

[Use In 2 = 0.7, R = 2 cal/K-mol]

TE0042

8. One mole of an ideal gas at 300 K is heated at constant volume (V_1) until it's temperature is doubled, then it is subjected to isothermal reversible expansion till it reaches the original pressure. Finally the gas is cooled at constant pressure till system reached to the half of original volume ($V_1/2$). Determine total work done ($|w|$) in cal. **[Use : ln2 = 0.70 ; R = 2 Cal/K.mol.]**

TE0043

9. How many times volume of diatomic gas should be expanded reversibly & adiabatically in order to reduce it's r.m.s velocity to half ?

TE0044

10. 0.5 mole each of two ideal gases A ($C_v = \frac{3}{2}R$) and B ($C_v = \frac{5}{2}R$) are taken in a container and expanded reversibly and adiabatically from V = 1 L to V = 4 L starting from initial temperature T = 300 K . Find $|\Delta H|$ for the process in cal.

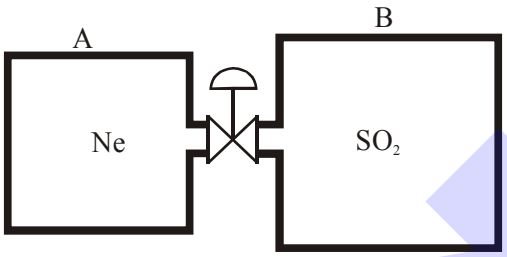
TE0045

11. 0.50 mol of an ideal gas initially at a temperature of 300 K and at a pressure of 2 atm is expanded isothermally in three steps. In each step, the pressure is dropped suddenly and held constant until equilibrium is re-established. The pressure at each of the three stages of expansion are 1.6, 1.2 and 1 atm. Calculate the work done ($|W|$) (in atm -litre) in this process.
[Use R = 0.08 atm litre/ mole. K]

TE0046

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol-TD\01-TC-TD-02 & 1.5_Eng (PC)\Eng\1-Thermodynamic-01\02_Ex.p65

12. Two rigid adiabatic vessels A and B which initially, contains two gases at different temperatures are connected by pipe line with valve of negligible volume. The vessel 'A' contains 2 moles Ne gas $\left(C_{p,m}=\frac{5}{2}R\right)$ at 300 K, vessel 'B' contains 3 moles of SO_2 gas ($C_{p,m}$ = 4 R) at 400 K. The volume of A & B vessel is 4 and 6 litre respectively. The final total pressure (in atm) when valve is opened and 12 Kcal heat supplied through it to vessels.
[Use : R = 2 cal/mol. K and R = 0.08 L. atm / mol K as per desire]



TE0047

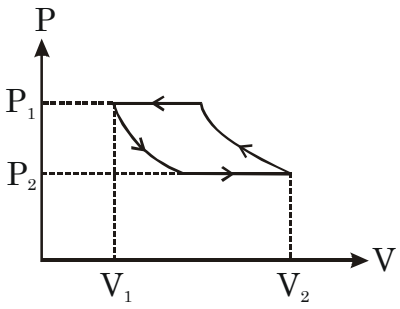
13. One mole of an ideal monoatomic gas initially at 1200 K and 64 atm is expanded to a final state at 300 K and 1.00 atm. To achieve change, a reversible path is constructed that involve an adiabatic expansion in the beginning followed by an isothermal expansion to the final state. Determine the magnitude of net work done by the gas, in cal. (R = 2 cal/K-mol. ℓn 2 = 0.7)

TE0048

14. An ideal gas is carried through a thermodynamic cycle.

P_1 = 2 bar $\quad V_1 = \frac{1}{\ell n10}$ litre

P_2 = 1 bar $\quad V_2 = \frac{8}{\ell n10}$ litre



Consisting of two isobaric and two isothermal processes. Calculate the net work in the entire cycle in litre bar. [log2 = 0.3]

TE0049

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

EXERCISE # O-I

BASIC

1. Out of boiling point (I), entropy (II), pH (III) and density (IV), intensive properties are

(A) I, II (B) I, II, III (C) I, III, IV (D) All of these

TE0050

2. What is true for a cyclic process [E = internal energy]

(A) W= 0 (B) $\Delta E = 0$ (C) $\Delta H = 0$ (D) B & C both

TE0129

3. The internal energy change when a system goes from state A to B is 40 kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy ? **[AIEEE-2003]**

(A) < 40 kJ (B) Zero (C) 40 kJ (D) > 40 kJ

TE0130

4. Representation the following observations in terms of proper IUPAC symbol -

(a) Heat absorbed by a system is 20 Joule.

(b) Work done by a system is 40 Joule.

(c) Work done on a system is 5 Joule.

(d) Heat given out by system is 50 Joule.

(A) + 20 J, –40 J , +5 J , – 50J (B) + 20 J, +40 J , +5 J , – 50J

(C) + 20 J, +40 J , –5 J , – 50J (D) – 20 J, +40 J , –5 J , + 50J

TE0131

5. Predict sign of work done in following reactions at constant pressure.

	Initial state		**Final state**
(i)	$H_2O(g)$	$\longrightarrow$	$H_2O(\ell)$
(ii)	$H_2O(s)$	$\longrightarrow$	$H_2O(g)$
(iii)	$H_2O(\ell)$	$\longrightarrow$	$H_2O(s)$
(iv)	$CaCO_3(s)$	$\longrightarrow$	$CaO(s) + CO_2(g)$

(A) **(i) + , (ii) –, (iii) – , (iv) –** (B) **(i) + , (ii) +, (iii) – , (iv) –**

(C) **(i) + , (ii) –, (iii) + , (iv) –** (D) **(i) + , (ii) +, (iii) – , (iv) –**

TE0132

6. A system has internal energy equal to U_1, 450 J of heat is taken out of it and 600 J of work is done on it. The final energy of the system will be -

(A) (U_1 + 150 J) (B) (U_1 + 1050 J) (C) (U_1 – 150 J) (D) (U_1 – 1050 J)

TE0133

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\e4-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

7. If work done by the system is 300 joule when 100 cal heat is supplied to it. The change in internal energy during the process is :-

(A) – 200 Joule (B) 400 Joule (C) 720 Joule (D) 120 Joule

TE0134

8. The work done by a system is 8J when 40J heat is supplied to it. The change in internal energy of the system during the process :

(A) 32 J (B) 40 J (C) 48 J (D) –32 J

TE0135

9. The work done during the expansion of a gas from a volume of 4 dm^3 to 6 dm^3 against a constant external pressure of 3 atm is -

(A) –608 J (B) +304 J (C) –304 J (D) –6 J

TE0136

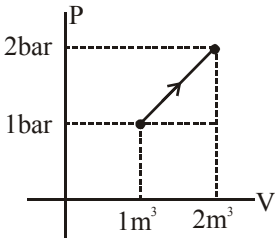
10. What is the change in internal energy when a gas is compressed from 377 ml to 177 ml under a constant pressure of 1520 torr, while at the same time being cooled by removing 124 J heat ?

[Take : (1L atm) = 100 J]

(A) –24 J (B) – 84 J (C) – 164 J (D) – 248 J

TE0051

11. What is ΔU for the process described by figure. Heat supplied during the process q = 200 kJ.

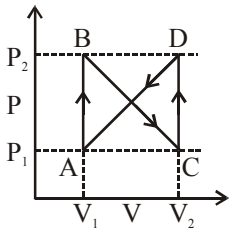


(A) +50 kJ (B) –50 kJ

(C) –150 kJ (D) + 150 kJ

TE0068

12. An ideal gas is taken around the cycle ABCDA as shown in figure. The net work done during the cycle is equal to :-



(A) Zero

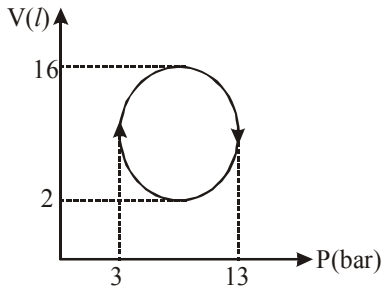
(B) Positive

(C) Negative

(D) We cannot predict

TE0067

13. Work (in kJ) in the following cyclic process is



(A) –11 (B) –11000 (C) 11 (D) 11000

TE0066

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-thermodynamic-01\02_Ex.p65

14. One mole of a gas absorbs 200J of heat at constant volume. Its temperature rises from 298 K to 308 K. The change in internal energy is :-

(A) 200 J (B) –200 J (C) $200 \times \frac{308}{298}$ J (D) $200 \times \frac{298}{308}$ J

TE0137

15. One mole of an ideal monoatomic gas expanded irreversibly in two stage expansion.

State-1 (8.0 bar, 4.0 litre, 300 K)
State-2 (2.0 bar, 16litre, 300 K)
State-3 (1.0 bar, 32 litre, 300 K)

Total heat absorbed by the gas in the process is :

(A) 116J (B) 40 J (C) 4000 J (D) None of these

TE0052

16. An ideal gas undergoes isothermal expansion from A(10 atm, 1*l*) to B(1 atm, 10 *l*) either by
(I) Infinite stage expansion or by
(II) First against 5 atm and then against 1 atm isothermally

Calculate $\frac{q_I}{q_{II}}$

(A) $\frac{1}{13 \times 2.303}$ (B) 13×2.303 (C) $\frac{13}{23.03}$ (D) $\frac{23.03}{13}$

TE0053

17. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0°C. As it does so, it absorbs 208 J of heat. The values of q and w for the process will be :- **[JEE-MAIN-2013]**

(R = 8.314 J/mol K) (ln 7.5 = 2.01)

(A) q = + 208 J, w = – 208 J (B) q = – 208 J, w = – 208 J
(C) q = – 208 J, w = + 208 J (D) q = + 208 J, w = + 208 J

TE0138

18. For an adiabatic process, which of the following relation must be correct -

(A) $\Delta U = 0$ (B) $P\Delta V = 0$ (C) $q = 0$ (D) $q = +W$

TE0055

19. A system containing ideal gas is expanded under adiabatic conditions. It shows :

(A) Temperature increase (B) Internal energy decrease
(C) Internal energy increase (D) None of these

TE0056

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1.5_{PC}\Eng\1-Thermodynamic-01\02_Ex.p65

20. A gas $\left(C_{v,m} = \frac{5}{2}R\right)$ behaving ideally was allowed to expand reversibly and adiabatically from 1 litre to 32 litre. It's initial temperature was 327° C. The molar enthalpy change (in J/mole) for the process is :-

(A) –1125 R (B) – 575 R (C) –1575 R (D) None of these

TE0057

21. Two moles of an ideal gas ($C_V = \frac{5}{2}R$) was compressed adiabatically against constant pressure of 2 atm, which was initially at 350 K and 1 atm pressure. The work involve in the process is equal to -

(A) 250 R (B) 300 R (C) 400 R (D) 500 R

TE0058

22. One mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C. If the work done by the gas in the process is 3 kJ, the final temperature will be equal to (C_V = 20 J/K mol)

(A) 100 K (B) 450 K (C) 150 K (D) 400 K

TE0058

23. What is the magnitude of work performed by one mole of an ideal gas when its volume increases eight times in irreversible adiabatic expansion if the initial temperature of the gas in 300 K ? C_V for the gas in 1.5 R. (R = 2 Cal / mol/ K)

(A) 900 Cal (B) 450 Cal (C) 675 Cal (D) 331.58 Cal

TE0060

24. A gas is expanded from volume V_1 to V_2 through three different process :

(a) Reversible adiabatic (b) Reversible isothermal

(c) Irreversible adiabatic

The correct statements is -

(A) $(T_f)_{\text{Reversible Isothermal}} > (T_f)_{\text{Reversible adiabatic}} > (T_f)_{\text{Irreversible adiabatic}}$

(B) $(T_f)_{\text{Reversible Isothermal}} > (T_f)_{\text{Irreversible adiabatic}} > (T_f)_{\text{Reversible adiabatic}}$

(C) $W_{\text{Reversible Isothermal}} > W_{\text{Irreversible adiabatic}} > W_{\text{reversible adiabatic}}$

(D) $(P_f)_{\text{Reversible Isothermal}} > (P_f)_{\text{Reversible adiabatic}} > (P_f)_{\text{Irreversible adiabatic}}$

TE0059

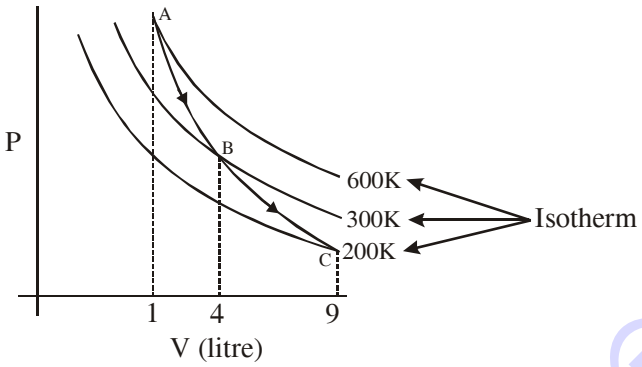
25. Two moles of an ideal gas expand spontaneously into vacuum. The work done is :–

(A) Zero (B) 2 J (C) 4 J (D) 8 J

TE0139

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4TD-01-TC-TD-02 &I.Eq.-(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

26. The given figure shows a polytropic process ABC for one mole of an ideal gas. Calculate the polytropic index (x) for the process-



(A) 1/2 (B) –1/2 (C) 3/2 (D) None

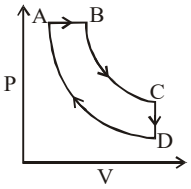
TE0065

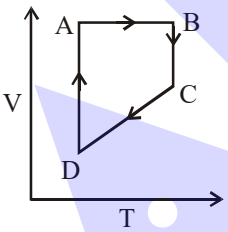
27. For a process which follows the equation $PV^3 = C$, the work done when one mole of ideal gas was taken from 1 atm to $2\sqrt{2}$ atm starting from initial temperature of 300 K is

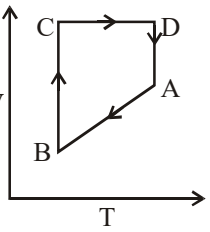
(A) 300 R (B) 150 R (C) 600 R (D) 900 R

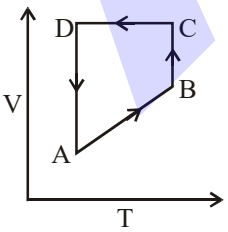
TE0061

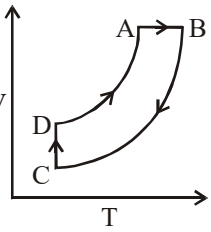
28. A cyclic process ABCD is shown in PV diagram for an ideal gas. which of the following diagram represents the same process ?



(A)


(B)


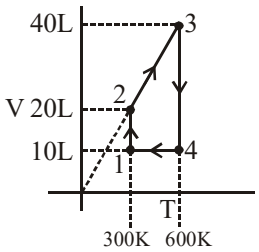
(C)


(D)


TE0063

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\e-TD-01-TC-TD-02 & 1.Eq_(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

29. What is the net work done (w) when 1 mole of monoatomic ideal gas undergoes in a process described by 1, 2, 3, 4 in given V–T graph



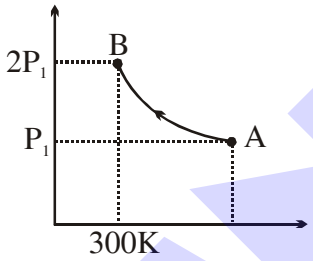
Use : R = 2cal/mole K

ln 2 = 0.7

(A) –600 cal (B) – 660 cal

(C) + 660 cal (D) + 600 cal

30. 2 moles of an ideal monoatomic gas is taken from state A to state B through a process AB in which PT = constant. The process can be represented on a P-T graph as follows:



Select the incorrect option(s):

(A) Heat evolved by the gas during process AB = 2100 R

(B) Heat absorbed by the gas during process AB = 2100 R

(C) $\Delta U = -900$ R

(D) W = 1200R

TE0064

31. For an ideal monoatomic gas during any process T = kV, find out the molar heat capacity of the gas during the process. (Assume vibrational degree of freedom to be active)

(A) $\frac{5}{2}R$ (B) 3R (C) $\frac{7}{2}$ (D) 4R

TE0070

32. An amount Q of heat is added to a monoatomic ideal gas in a process in which the gas perfomrs a work Q/2 on its surrounding. The molar heat capacity of gas (in cal/K-mol) for the process is.

(A) 3R (B) 5R (C) 4R (D) 2R

TE0071

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4TD-01-TC-TD-02 &1.Eq.-IPC\Eng\1-Thermodynamic-01\02_Ex.p65

EXERCISE # O-II

❑ ONE OR MORE THAN ONE MAY BE CORRECT :

1. An insulated container of gas has two chambers separated by an insulating partition. One of the chambers has volume V_1 and contains ideal gas at pressure P_1 and temperature T_1. The other chamber has volume V_2 and contains same ideal gas at pressure P_2 and temperature T_2. If the partition is removed without doing any work on the gas, the final equilibrium temperature of the gas in the container will be :-

(A) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_2+P_2V_2T_1}$ (B) $\frac{P_1V_1T_1+P_2V_2T_2}{P_1V_1+P_2V_2}$ (C) $\frac{P_1V_1T_2+P_2V_2T_1}{P_1V_1+P_2V_2}$ (D) $\frac{T_1T_2(P_1V_1+P_2V_2)}{P_1V_1T_1+P_2V_2T_2}$

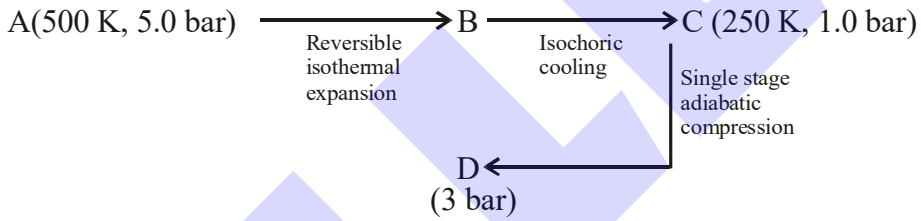
TE0072

2 The heat capacity of liquid water is 75.6 J/mol.K, while the enthalpy of fusion of ice is 6.0 kJ/mol. What is the smallest number of ice cubes at 0°C, each containing 9.0 g of water, needed to cool 500 g of liquid water from 20°C to 0°C ?

(A) 1 (B) 7 (C) 14 (D) 252

TE0073

3. Two moles of an ideal gas ($C_{v,m} = \frac{3}{2}R$) is subjected to following change of state



The correct statement is/are

(A) The pressure at B is 2.0 bar (B) The temperature at D is 450 K

(C) $\Delta H_{CD} = 1000$ R (D) $\Delta U_{BC} = 375$ R

TE0074

4. A real gas is subjected to an adiabatic process from (2 bar, 40 lit., 300 K) to (4 bar, 30 lit., 300 K) against a constant pressure 4 bar the enthalpy change for the process is

(A) Zero (B) 6000 J (C) 8000 J (D) 80 J

TE0075

5. Which of the following statement is/are correct ?

(A) Enthalpy can be written as H = f(V,T) for a substance (no physical or chemical change)

(B) Absolute value of enthalpy can not be determined

(C) The heat absorbed during the isothermal expansion of an ideal gas against vacuum is zero

(D) During compression of an ideal gas at constant pressure the temperature of gas decreases.

TE0076

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol-TD-01-TC-TD-02 & I.Eq. (PC)\Eng\1-Thermodynamic-01\02_Ex.p65

6. Choose the correct statement(s)

(A) During adiabatic expansion of an ideal gas, magnitude of work obtained is equal to ΔH of gas.

(B) For same change in temperature of ideal gas through adiabatic process, magnitude of W will be same in reversible as well as irreversible process

(C) During an adiabatic reversible expansion of an ideal gas, temperature of the system increases.

(D) For same change in volume of ideal gas, ΔU is less in adiabatic expansion than in isobaric expansion if expansion starts with same intial state.

TE0077

7. Choose the correct statement(s) among the following -

(A) Internal energy of H_2O remains constant during conversion of liquid into its vapour at constant temperature.

(B) During fusion of ice into water, enthalpy change and internal energy change are almost same at constant temperature

(C) Molar heat capicity of gases are temperature dependent

(D) During boiling of H_2O at 1 atm, 100ºC, average kinetic energy of H_2O molecules increases.

TE0078

8. Suppose that the volume of a certain ideal gas is to be doubled by one of the following processes :

(A) isothermal expansion (B) adiabatic expansion

(C) free expansion in insulated condition (D) expansion at constant pressure.

If E_1,E_2,E_3 and E_4 respectively are the changes in average kinetic energy of the molecules for the above four processes, then-

(A) $E_2 = E_3$ (B) $E_1 = E_3$ (C) $E_1 > E_4$ (D) $E_4 > E_3$

TE0079

❑ **Assertion / reason :**

9. **Statement-1 :** There is no change in enthalpy of an ideal gas during compression at constant temperature.

Statement-2 : Enthalpy of an ideal gas is a function of temperature and pressure.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

TE0080

10. **Statement-1 :** Due to adiabatic expansion, the temperature of an ideal gas always decreases.

Statement-2 : For an adiabatic process, $\Delta U = w$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

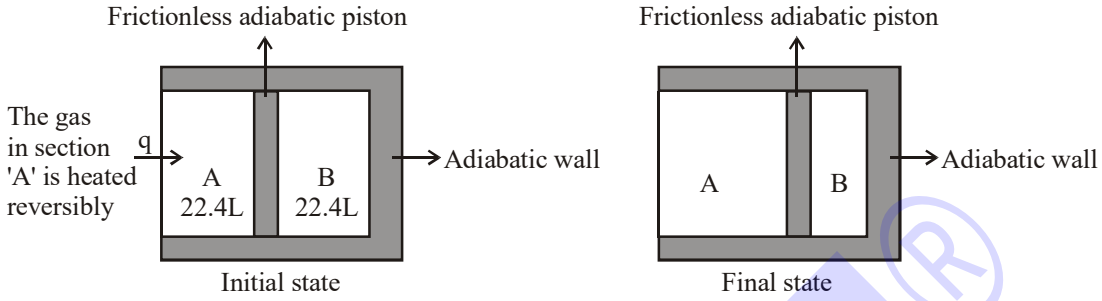
(D) Statement-1 is false, statement-2 is true.

TE0081

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & I. Eq.-(PC)\Eng\1-thermodynamic-01\02_Ex.p65

Paragraph for Q.11 to Q.13

A cylindrical container of volume 44.8 litres is containing equal no. of moles (in integer no.) of an ideal monoatomic gas in two sections A and B separated by an adiabatic frictionless piston as shown in figure. The initial temperature and pressure of gas in both section is 27.3K and 1 atm. Now gas in section 'A' is slowly heated till the volume of section B becomes $(1/8)^{th}$ of initial volume.



Given : R = 2 cal/mol-K, $C_{v,m}$ of monoatomic gas = $\frac{3}{2}$R,

At 1 atm & 0°C ideal gas occupy 22.4 litre.

11. What will be the final pressure in container B.

(A) 2 atm (B) 8 atm (C) 16 atm (D) 32 atm

TE0082

12. Final temperature in container A will be

(A) 1638 K (B) 6988 K (C) 3274 K (D) 51 K

TE0083

13. Change in enthalpy for section A in Kcal

(A) 48.3 (B) 80.53 (C) 4.83 (D) 8.05

TE0084

Paragrah for Q.14 to 16

As a result of the isobaric heating by $\Delta T = 72K$, one mole of a certain ideal gas obtains an amount of heat Q = 1.60 kJ.

14. The work performed by the gas is -

(A) 8.60 kJ (B) 0.60 kJ (C) 16.60 kJ (D) 4.60 kJ

TE0085

15. The increment of its internal energy (in kJ) is

(A) 1.0 (B) 1.6 (C) 2.2 (D) 2.0

TE0086

16. The value of γ for the gas is -

(A) 0.6 (B) 1.4 (C) 1.6 (D) 1.5

TE0087

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol-TD-01-TC-TD-02 & 1.5_{PC}\Eng\1-Thermodynamic-01\02_Ex.p65

MATCH THE COLUMN :

17. Match Column-I with Column-II

	Column–I (Ideal Gas)		**Column-II (Related equations)**
(A)	Reversible isothermal process	(P)	$W = 2.303\ n\ RT\ \log(P_2/P_1)$
(B)	Reversible adiabatic process	(Q)	$W = nC_{V,m}\ (T_2 - T_1)$
(C)	Irreversible adiabatic process	(R)	$W = -2.303nRT\ \log(V_2/V_1)$
(D)	Irreversible isothermal process	(S)	$W = -\int_{V_i}^{V_f} P_{ext.}\,dV$

TE0088

18. Match the column

	Column–I		**Column–II**
(A)	Isothermal vaporisation of water at 100°C & 1 atm	(P)	$\Delta T = 0$
(B)	Isothermal reversible expansion of an ideal gas	(Q)	$\Delta U = 0$
(C)	Adiabatic free expansion of ideal gas	(R)	$\Delta H = 0$
(D)	Isochoric heating of an ideal gas	(S)	$q = 0$
		(T)	$w = 0$

TE0089

MATCH THE LIST :

19. Match the following, and select the correct code :-

	List-I		**List-II**
(P)	Isothermal process (reversible)	(1)	$q = 2.303\ nRT \log \frac{P_1}{P_2}$
(Q)	Adiabatic process (reversible)	(2)	$PV^{\gamma} = \text{const.}$
(R)	Isochoric process	(3)	$q = P_{ext.}\ (V_2 - V_1)$
(S)	Isothermal process (irreversible)	(4)	Area under P-V curve is zero

Code :

	P	**Q**	**R**	**S**
(A)	2	1	4	3
(B)	1	2	4	3
(C)	2	1	3	4
(D)	1	2	3	4

TE0090

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol4TD-01-TC-TD-02 &1.Eq.-IPC\Eng\1-thermodynamic-01\02_Ex.p65

20. The figures given below depict different processes for a given amount of an ideal gas.

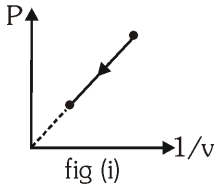
(A) 

fig (i)

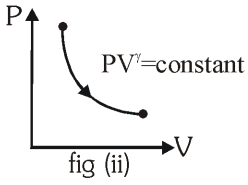
(B) 

fig (ii)

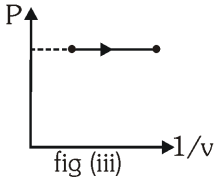
(C) 

fig (iii)

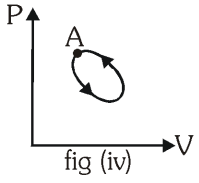
(D) 

fig (iv)

List-I		List-II	
(P)	In Fig (i)	(1)	Net heat is absorbed by the system
(Q)	In Fig (ii)	(2)	Net work is done on the system
(R)	In Fig (iii)	(3)	Net heat is rejected by the system
(S)	In Fig (iv)	(4)	Net work is done by the system
		(5)	Net internal energy change is zero

Code:

	P	Q	R	S
(A)	1, 4, 5	4	2, 3	2. 3.5
(B)	1, 4, 5	4	2, 3	2, 3
(C)	1, 2, 4	2	1, 4	2, 3
(D)	1, 2	2	1, 4	3, 5

TE0091

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02&15_(PC)\Eng\1-Thermodynamic-01\02_Ex.pmd5

EXERCISE # J-MAIN

1. A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0 mL to 375 mL at a constant temperature of 37.0°C. As it does so, it absorbs 208 J of heat. The values of q and w for the process will be :- **[JEE-MAIN-2013]**

(R = 8.314 J/mol K) (ln 7.5 = 2.01)

(1) q = + 208 J, w = – 208 J

(2) q = – 208 J, w = – 208 J

(3) q = – 208 J, w = + 208 J

(4) q = + 208 J, w = + 208 J

TE0092

2. Which of the following statements/relationships is not correct in thermodynamic changes ?

(1) $q = -nRT\,\ell n\frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas) **[JEE-MAINS(online)-2014]**

(2) For a system at constant volume, heat involved merely changes to internal energy.

(3) $w = -nRT\ln\frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

(4) $\Delta U = 0$ (isothermal reversible expansion of a gas)

TE0093

3. A gas undergoes change from state A to state B. In this process , the heat absorbed and work done by the gas is 5 J and 8 J, respectively. Now gas is brought back to A by another process during which 3 J of heat is evolved. In this reverse process of B to A:-

[JEE-MAINS(online)-2017]

(1) 10 J of the work will be done by the surrounding on gas.

(2) 6 J of the work will be done by the surrounding on gas.

(3) 10 J of the work will be done by the gas.

(4) 6 J of the work will be done by the gas.

TE0094

4. The enthalpy change on freezing of 1 mol of water at 5°C to ice at –5°C is :

(Given $\Delta_{fus}H$ = 6 kJ mol^{-1} at 0°C, $C_P(H_2O, l)$ = 75.3 J mol^{-1} K^{-1}, $C_P(H_2O, s)$ = 36.8 J mol^{-1} K^{-1})

[JEE-MAINS(online)-2017]

(1) –6.56 kJ mol^{-1} (2) –5.81 kJ mol^{-1} (3) –6.00 kJ mol^{-1} (4) –5.44 kJ mol^{-1}

TE0095

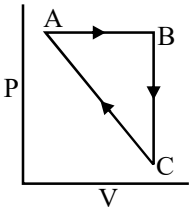
5. ΔU is equal to **[JEE-MAINS(online)-2017]**

(1) Isochoric work (2) Isobaric work (3) Adiabatic work (4) Isothermal work

TE0096

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

6. An ideal gas undergoes a cyclic process as shown in figure. **[JEE-MAINS(online)-2018]**



$\Delta U_{BC} = -5$ kJ mol^{-1}, $q_{AB} = 2$ kJ mol^{-1}

$W_{AB} = -5$ kJ mol^{-1}, $W_{CA} = 3$ kJ mol^{-1}

Heat absorbed by the system during process CA is :-

(1) 18 kJ mol^{-1} (2) +5 kJ mol^{-1} (3) –5 kJ mol^{-1} (4) –18 kJ mol^{-1}

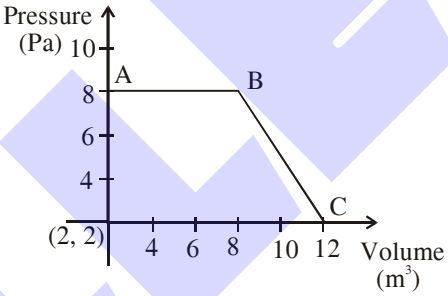
TE0097

7. An ideal gas undergoes isothermal compression from 5 m^3 to 1 m^3 against a constant external pressure of 4 Nm^{-2}. Heat released in this process is used to increase the temperature of 1 mole of Al. If molar heat capacity of Al is 24 J mol^{-1} K^{-1}, the temperature of Al increases by **[JEE-MAINS(online)-2019]**

(1) $\frac{3}{2}$K (2) $\frac{2}{3}$K (3) 1 K (4) 2 K

TE0099

8. The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is _____ **[JEE-MAINS(online)-2020]**

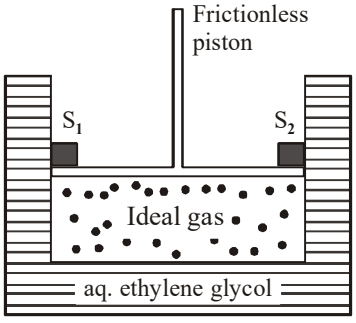


TE0100

9. At constant volume, 4 mol of an ideal gas when heated from 300 K to 500K changes its internal energy by 5000 J. The molar heat capacity at constant volume is _____. **[JEE-MAINS(online)-2020]**

TE0101

10. A cylinder containing an ideal gas (0.1 mol of 1.0 dm^3) is in thermal equilibrium with a large volume of 0.5 molal aqueous solution of ethylene glycol at its freezing point. If the stoppers S_1 and S_2 (as shown in the figure) are suddenly withdrawn, the volume of the gas in litres after equilibrium is achieved will be____. (Given, K_f (water) = 2.0 K kg mol^{-1}, R = 0.08 dm^3 atm K^{-1} mol^{-1}) **[JEE-MAINS(online)-2020]**



TE0102

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD\01-TC-TD-02 & 1.5_{PC}\Eng\1-Thermodynamic-01\02_Ex.p65

11. A system does 200 J of work and at the same time absorbs 150 J of heat. The magnitude of the change in internal energy is _____ J. (Nearest integer) **[JEE-MAINS(online)-2021]**

TE0140

12. Five moles of an ideal gas at 293 K are expanded isothermally from an initial pressure of 2.1 MPa to 1.3 MPa against at constant external pressure 4.3 MPa. The heat transferred in this process is ____kJ.

(Rounded-off to the nearest integer)

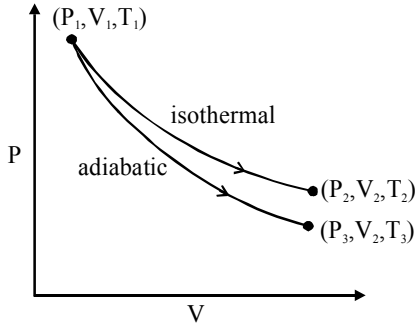
[Use R = 8.314 J $mol^{-1}K^{-1}$] **[JEE-MAINS(online)-2021]**

TE0141

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-thermodynamic-01\02_Ex.p65

EXERCISE # J-ADVANCED

1. The reversible expansion of an ideal gas under adiabatic and isothermal conditions is shown in the figure. Which of the following statement(s) is (are) correct ? **[JEE 2012]**

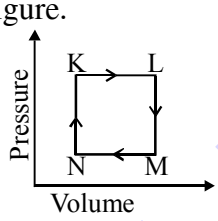


(A) $T_1 = T_2$ (B) $T_3 > T_1$

(C) $w_{isothermal} > w_{adiabatic}$ (D) $\Delta U_{isothermal} > \Delta U_{adiabatic}$

Paragraph for Question 2 and 3

A fixed mass 'm' of a gas is subjected to transformation of states from K to L to M to N and back to K as shown in the figure. **[JEE 2013]**



TE0103

2. The pair of isochoric processes among the transformation of states is

(A) K to L and L to M (B) L to M and N to K

(C) L to M and M to N (D) M to N nd N to K

TE0104

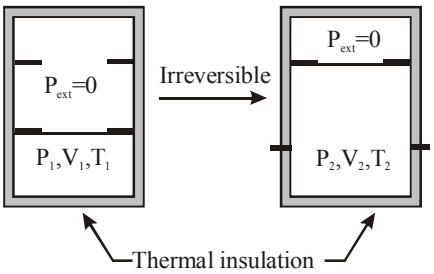
3. The succeeding operations that enable this transformation of states are

(A) Heating, cooling, heating, cooling (B) cooling, heating, cooling, heating

(C) Heating, cooling, cooling, heating (D) Cooling, heating, heating, cooling

TE0105

4. An ideal gas in thermally insulated vessel at internal pressure = P_1, volume = V_1 and absolute temperature = T_1 expands irrversibly against zero external pressure, as shown in the diagram. The final internal pressure, volume and absolute temperature of the gas are P_2, V_2 and T_2, respectively. For this expansion, **[JEE 2014]**



(A) q = 0 (B) $T_2 = T_1$ (C) $P_2V_2 = P_1V_1$ (D) $P_2V_2^{\gamma} = P_1V_1^{\gamma}$

TE0106

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\4-TD-01-TC-TD-02 & 1.5_{PC}\Eng\1-Thermodynamic-01\02_Ex.p65

5. An ideal gas is expanded from (p_1, V_1, T_1) to (p_2, V_2, T_2) under different conditions. The correct statement(s) among the following is(are) :- **[JEE 2017]**

(A) The work done on the gas is maximum when it is compressed irreversibly from (p_2, V_2) to (p_1, V_1) against constant pressure p_1

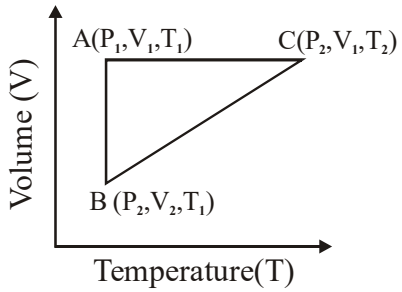
(B) The work done by the gas is less when it is expanded reversibly from V_1 to V_2 under adiabatic conditions as compared to that when expanded reversibly from V_1 to V_2 under isothermal conditions.

(C) The change in internal energy of the gas (i) zero, if it is expanded reversibly with $T_1 = T_2$, and (ii) positive, if it is expanded reversibly under adiabatic conditions with $T_1 \neq T_2$

(D) If the expansion is carried out freely, it is simultaneously both isothermal as well as adiabatic

TE0107

6. A reversible cyclic process for an ideal gas is shown below. Here, P , V and T are pressure , volume and temperature , respectively. The thermodynamic parameters q, w, H and U are heat, work, enthalpy and internal energy, respectively. **[JEE 2018]**



The correct option(s) is (are)

(A) $q_{AC} = \Delta U_{BC}$ and $w_{AB} = P_2 (V_2 - V_1)$

(B) $w_{BC} = P_2 (V_2 - V_1)$ and $q_{BC} = \Delta H_{AC}$

(C) $\Delta H_{CA} < \Delta U_{CA}$ and $q_{AC} = \Delta U_{BC}$

(D) $q_{BC} = \Delta H_{AC}$ and $\Delta H_{CA} > \Delta U_{CA}$

TE0108

7. In thermodynamics the P-V work done is given by **[JEE 2020]**

$$w = -\int dV\, P_{ext}$$

For a system undergoing a particular process, the work done is ,

$$w = -\int dV \left(\frac{RT}{V-b} - \frac{a}{V^2} \right)$$

This equation is applicable to a

(A) System that satisfies the van der Waals equation of state.

(B) Process that is reversible and isothermal.

(C) Process that is reversible and adiabatic.

(D) Process that is irreversible and at constant isothermal.

TE0142

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

ANSWER KEY

EXERCISE # S-I

1. 21.00 **2.** (7.00) **3.** (–65.00 , 20.00 & –45.00)

4. (–4.00) **5.** (a) 170.00 (b) 10.00 **6.** (a) 60.00 (b) –70.00 (c) 50.00 , 10 .00

7. (i) 9, 3, 2, 4 (ii) 9, 3, 3, 3 (iii) 3, 3, 0, 0 (iv) 12, 3, 3, 6

8. (8250.00) **9.** (420.00) **10.** (9.00) **11.** (4.00)

12. (3.50) **13.** $\Delta H = 1000.00$, $\Delta U = 970.00$

14. 1440.00 , (1440.03 to 1440.04) **15.** –3.90 to –4.00 **16.** (3.50)

17. (–900.00)

18. Irreversible : 92.36, –92.36, 0.00 , 0.000 ; reversible : 172.41, –172.41, 0.00, 0.00

19. (13.00) **20.** (i) 18.42 ; (ii) 72.00 ; (iii) 40.00

21. –3.00 **22.** (140.00) **23.** (0.00, –300.00 , –420.00, –300.00)

24. –562.50, –1.20 **25.** $T_2 = 100.00$; –5.01 or –5.02

26. 0, 4.12 , 4.12 , 5.37 , 11.8 , 5.22 **27.** (500.00)

28.

Table-1			
State	P	V	T
1	1 atm	22.4	273
2	1	44.8	546
3	0.5	44.8	273

Step	Name of process	q	w	ΔE	ΔH
A	Isobaric	1365	–546	819	1365
B	Isochoric	– 819	0	– 819	– 1365
C	Isothermal	– 382.2	382.2	0	0
	Cyclic	163.8	– 163.8	0	0

29.

Table-1			
State	P	V	T
1	1 atm	22.4	273
2	2 atm	22.4	546
3	1 atm	44.8	546

Step	Name of process	q	w	ΔE	ΔH
A	Isochoric	819.00	0	819.00	1365.00
B	Isothermal	764.40	-764.40	0	0
C	Isobaric	-1365.00	546.00	–819.00	–1365.00
	Overall	218.40	-218.40	0	0

30. (a) 244 , 2436 , (b) 0.00 , 3264, –3264 **31.** 7.00 **32.** –800.00 **33.** –0.25

34. (a) 3.67 (b) 2566.67 (c) –466.67 (d) 2100.00

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Vol-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-Thermodynamic-01\02_Ex.p65

EXERCISE # S-II

1. (a) 600 K, (b) 3 kcal, 1.68 kcal, –1.80 kcal, –1.68 kcal (c) w = –1.20 kcal

2. ΔU & ΔH = 0; w = 623.55 J ; q = –623.55 J

3. –16.67 to –16.68

4. (i) w = –8400 cal , q = 8400cal , ΔU = ΔH = 0.

(ii) w = –2700 cal , q = 0cal , ΔU = – 2700, ΔH = –4500 cal

(iii) w = 0 cal , q = 0 cal , ΔU = 0 , ΔH = 0.

(iv) w = – 1395 cal q = 0, ΔU = –1395 cal , ΔH = –2325 cal

(v) w = – 2325 cal q = 2325 cal , ΔU = 0 , ΔH = 0

5. ΔU = 501 J; ΔH = 99.50 kJ **6.** 100 KJ/mole **7.** –1200 cal

8. 60 cal **9.** (32) **10.** (900 calorioes) **11.** 7.40 atm litre

12. 35 atm **13.** 3120 **14.** (1.80 bar-litre)

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	D	B	A	A	A	D	A	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	C	A	C	D	A	C	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	C	D	B	A	C	B	C	C	B
Que.	31	32								
Ans.	A	A								

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	A,B,C	C	A,B,C,D	B,D	B,C	B,D	C	D
Que.	11	12	13	14	15	16	17			
Ans.	D	A	B	B	A	C	A-P,R,S	B-Q,S	C-Q,S	D-S
Que.	18				19	20				
Ans.	A-P	B-P,Q,R	C-P,Q,R,S,T	D-T	B	A				

EXERCISE # J-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	1	2	1	3	2	2	48.00	6.25	2.17 to 2.23
Que.	11	12								
Ans.	50	15								

EXERCISE # (J-ADVANCED)

Que.	1	2	3	4	5	6	7
Ans.	A,D	B	C	A,B,C	A,B,D	B,C	A,B,C

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\s4-TD-01-TC-TD-02 & I.Eq.-(PC)\Eng\1-thermodynamic-01\02_Ex.p65